SEICHO

Tool Lifetime Management System Application Requirement

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INTRODUCTION

This document outlines the functional requirements for the development of the **Rosenberger Tools Lifetime Management System (TLMS)**, a centralized platform for managing machine tools and monitoring their consumption and remaining lifespan for **Rosenberger India Group**.

SEICHO is pleased to collaborate with Rosenberger India Group in delivering a scalable and configurable solution that streamlines tool lifecycle management, improves data accuracy, and standardizes key operational processes. The platform is designed to digitize tool onboarding, track tool consumption across machines and shifts, automatically calculate remaining tool lifespan, and support effective monitoring and administrative control.

The **Rosenberger Tools Lifetime Management System** will provide a structured framework for managing tool and machine master data, recording shift-wise consumption, handling missed consumption entries and administrative overrides, monitoring tool health through dashboards, and exporting relevant operational data. Role-based access for Operators, Setters, Supervisors, and Administrators will support controlled access to information and functions, while centralized master data management and Excel upload capabilities will help maintain consistent and accurate records.

The following sections describe the functional scope, business roles, key processes, administrative responsibilities, and proposed modules of the Rosenberger Tools Lifetime Management System.

Roles & Responsibilities

The Tool Lifetime Management System (TLMS) involves multiple stakeholders, each responsible for specific activities throughout the tool lifecycle. The table below outlines the primary roles and their associated responsibilities within the Rosenberger Tools Lifetime Management System.

Role	Responsibilities
Operator	Records tool consumption at the end of each shift for the tools used during production. Ensures consumption information is entered accurately against the relevant Tool Identifier and machine.
Setter	Uses the system to access relevant tool and machine information and make necessary updates to the tools information and perform activities permitted by the assigned role.
Shift Supervisor	Records missed consumption on behalf of an operator when required. Ensures the correct consumption details are captured and the system reflects the updated remaining tool lifespan.
Administrator	Manages tool, machine, and user master data; configures shifts; assigns operators to shifts and supervisors to shifts; performs authorized consumption updates and zero-consumption operations; manages tool replacement records; uploads master data through Excel; and accesses dashboards and data exports according to permissions.

Functional Scope

The Rosenberger Tools Lifetime Management System (TLMS) is designed to provide a centralized solution for managing the tool lifecycle, tracking shift-wise consumption, monitoring remaining tool lifespan, and supporting administrative control. The platform comprises the following functional modules:

Module	Description
Tool Onboarding and Registration	Enables authorized users to register tools by capturing the Tool Identifier, associated machine details, recipe number, item name, maximum lifespan, description, and other relevant information required for tool lifecycle tracking.
Machine and Tool Master Data Management	Enables administrators to maintain machine and tool lists, manage tool-to-machine associations, update relevant master data, and maintain historical records when machines or tools become inactive.
Shift-wise Tool Consumption Recording	Enables operators to record tool consumption at the end of each shift against the relevant Tool Identifier and machine. The system automatically updates the remaining tool lifespan based on the recorded consumption.
Missed Consumption Management	Enables the relevant Shift Supervisor to record missed consumption on behalf of an operator when required. The system captures the supervisor identity and timestamp and updates the remaining tool lifespan accordingly.
Administrative Consumption Updates	Enables administrators to update consumption for a selected date and apply a common consumption value to one or more selected tools. The system recalculates the remaining lifespan based on the updated records.
Bulk Zero-Consumption Management	Enables administrators to mark consumption as zero for selected tools on a specified date, supporting exceptional production scenarios and corrections to consumption records according to agreed business rules.

Module	Description
Tool Replacement Management	Enables authorized users to record tool replacements while retaining the historical records of the old tool. The replacement tool starts with zero recorded consumption, and its remaining lifespan is initialized to its maximum lifespan.
Dashboard and Tool Health Monitoring	Provides a consolidated dashboard showing the total number of tools and their health status, such as Healthy, Moderate, and Critical. Includes graphical visualizations of tool consumption, lifespan, and other relevant operational data.
Role-Based Access and Data Views	Provides role-based access for Operators, Setters, Shift Supervisors, and Administrators, allowing each role to access the information and functions relevant to its responsibilities.
Data Export and Reporting	Enables authorized users to export agreed tool, machine, consumption, and lifespan data using applicable filters and access permissions.
User, Shift, and Supervisor Management	Enables administrators to maintain user details, configure shifts and working days, assign operators to shifts, and maintain Shift Supervisor assignments.
Excel-Based Master Data Upload	Enables administrators to upload user, machine, and tool master data through Excel files, with validation of required fields, duplicate records, associations, and reporting of accepted and rejected rows.

Detailed Functional Requirements

This section describes the detailed functional requirements of the Rosenberger Tools Lifetime Management System (TLMS). It defines the system capabilities, business rules, user activities, and administrative functions required to support the complete tool lifecycle, from tool onboarding and consumption tracking to lifespan monitoring, tool replacement, reporting, and master data management.

The following subsections describe the functional requirements for each area of the system.

1. Tools Onboarding

1.1 Machine and Location Management

- The system shall allow the administrator to maintain the list of machines used in production.
- Each machine shall have a machine ID and location.
- Users shall be able to select an existing machine when associating a tool with a machine.
- The system shall prevent unintended duplicate machine records.

1.2 Tool Registration

- The system shall allow authorized users to register and maintain tool information.
- The tool master record shall capture the following information:
    - Tool ID **or** Product Code, as the applicable identifier
    - Machine association
    - Machine ID
    - Recipe number
    - Tool Cost
    - Item name
    - Maximum lifespan
    - Tool status
- The system shall validate the mandatory information before saving a tool record.
- The system shall support the agreed tool identification and duplicate-handling rules.
- The system shall allow authorized users to update tool information as required.

1.3 Tool Associations

- The system shall support associating a tool with multiple machines where required by the production process.
- The system shall maintain the relevant machine, recipe, and product associations for each tool.
- The association rules for tools, recipes, and products shall follow the agreed operational requirements.

1.4 Initial Lifespan Setup

- The system shall establish the initial remaining lifespan based on the tool's configured maximum lifespan.
- The system shall support the agreed process for registering tools that have already been used before onboarding.
- The system shall display the configured maximum and remaining lifespan to authorized users.

1.5 Tool Status Management

- The system shall allow authorized users to maintain tool status.
- The permitted statuses shall be defined according to operational requirements.
- The system shall apply the agreed rules for consumption recording and tool replacement based on tool status.

2. Consumption Update

2.1 Shift-Wise Consumption Recording

- The system shall allow operators to record tool consumption after each production shift.
- Each consumption entry shall identify the relevant:
  - Tool.
  - Machine.
  - Production date.
  - Shift.
  - Consumption quantity.
- The system shall record the user responsible for submitting the consumption entry and the entry timestamp.
- The system shall support the three standard production shifts on working days.

- The system shall validate consumption entries according to the agreed business rules, including permissible quantities, valid associations, and duplicate-entry restrictions.

2.2 Automatic Lifespan Calculation

- The system shall automatically update the remaining lifespan when a valid consumption entry is saved.
- The calculation shall follow the rule:

  Remaining Lifespan = Previous Remaining Lifespan – Recorded Consumption

- For example, if a tool has 100 units of remaining lifespan and consumption of 10 units is recorded, the updated remaining lifespan shall be 90 units.
- The system shall reflect the updated lifespan in the relevant tool records, user views, and dashboard information.
- Where multiple valid consumption entries exist, the remaining lifespan shall reflect the cumulative recorded consumption.

2.3 Tools Approaching the End of Their Lifespan

- The system shall display the remaining lifespan of tools to authorized users.
- The system shall classify tools into the following categories:
  - **Healthy:** Sufficient remaining lifespan for continued use.
  - **Moderate:** Approaching the agreed attention threshold.
  - **Critical:** Reaching or approaching the agreed replacement threshold.
- The classification thresholds shall be defined according to the agreed business rules.
- The system shall handle tools that reach zero remaining lifespan according to the agreed rules for further use, recording, and replacement.

2.4 Missed Consumption Entries

- If an operator fails to record consumption for a production shift, the system shall allow the operator's immediate shift supervisor to enter the missing consumption on the operator's behalf.
- The supervisor shall be able to identify the relevant production date, shift, machine, and tool, and enter the applicable consumption quantity.
- The system shall record the identity of the supervisor who entered the missing consumption and the date and time of the entry.
- Once the missing consumption is saved, the system shall automatically recalculate the affected tool's remaining lifespan.

2.5 Incorrect Consumption Entries

- The system shall support correction of incorrectly recorded consumption by authorized users.
- The system shall recalculate the affected tool's remaining lifespan when a correction is saved.
- Corrections shall follow the agreed permission and approval rules.
- The system shall retain the agreed history of consumption corrections.

2.6 Sunday and Exceptional Production

- Sunday shall be treated as a non-working day under the normal operating schedule.
- The system shall allow the administrator to perform bulk consumption updates by:
    - Selecting a production date.
    - Selecting one or more tools.
    - Entering a common consumption value for the selected tools.
- The system shall apply the specified consumption value to each selected tool for the selected date.
- The system shall automatically recalculate the remaining lifespan of each affected tool.
- Before saving the bulk update, the system shall display the selected date, tools, and consumption value for confirmation.
- This functionality shall support exceptional Sunday production and other situations where the same consumption value must be applied to multiple tools for a specific date.
- The handling of existing consumption entries for the selected date and tools shall follow the agreed business rules.

3. Admin Override Functionality

3.1Consumption Override and Correction

- The system shall provide authorized administrators with the ability to correct or override consumption records.
- The system shall validate revised consumption values and recalculate the affected tool's remaining lifespan.
- The system shall follow the agreed rules for correction reasons, approvals, and permissions.
- The system shall retain the agreed history of changes, including the original and revised consumption values and the user responsible for the change.

3.2 Bulk Zero-Consumption Operation

- The system shall allow the administrator to select one or more tools and a specific production date.
- The administrator shall be able to mark consumption as zero for the selected tools on the selected date.
- The system shall display the selected tools, date, and affected records before the operation is saved.
- The system shall update the affected consumption records according to the agreed business rules.
- The system shall recalculate the remaining lifespan of each affected tool based on the resulting valid consumption records.
- The system shall retain the agreed history of bulk operations.

3.3 Tool Replacement

- The system shall support recording the replacement of a tool when its lifespan is exhausted or replacement is otherwise required.
- The replacement tool shall start with a consumption counter of zero.
- The replacement tool's remaining lifespan shall be established from its applicable maximum lifespan, subject to agreed initial-use adjustments.
- The system shall maintain the distinction between the old tool and its replacement.
- The system shall retain the agreed consumption and lifespan history of the old tool.
- The system shall maintain the relevant machine, recipe, and product associations for the replacement tool.
- The system shall support the agreed process for updating the old tool's status following replacement.

4. Dashboard, Role-Based Views and Export

4.1 Role-Based Views

- The system shall provide appropriate views for:
  - Operator.
  - Admin.
  - Setter.
  - Supervisor.
- Each role shall have access to the tool information, consumption details, and lifespan data relevant to its operational responsibilities.
- The system shall control access to activities such as tool registration, consumption entry, consumption correction, tool replacement, bulk operations, master data maintenance, and data export.
- Role-specific permissions shall follow the agreed operational responsibilities.

4.2 Dashboard Overview

- The dashboard shall display the total number of tools within the user's permitted scope.
- The dashboard shall display tool counts by lifespan category:
  - Healthy.
  - Moderate.
  - Critical.
- The dashboard shall provide a clear visual summary of the current tool lifespan condition.
- The system shall apply the agreed thresholds when classifying tools.
- Dashboard figures shall remain consistent with the latest valid consumption records and remaining lifespan balances.

4.3 Consumption Graphs and Data Visualization

- The dashboard shall provide graphical representations of relevant tool consumption data.
- The system shall display consumption trends and summaries based on the data available in the system.
- Relevant visualizations may include consumption by date, shift, machine, tool, or other applicable operational categories.
- The dashboard shall support graphical comparison of tool lifespan conditions and consumption patterns where relevant data is available.
- The system shall provide other relevant graphs and visual summaries based on available tool, machine, shift, consumption, and lifespan data.

- Users shall be able to access relevant tool-level details from dashboard visualizations, subject to their permissions.
- The dashboard shall present information in a clear format that supports operational monitoring and identification of tools requiring attention.

4.4 Data Export

- The system shall provide export functionality for authorized users.
- Exportable information shall include the agreed tool, machine, recipe, product, consumption, remaining lifespan, and tool status data.
- Users shall be able to apply agreed filters, such as production date, machine, tool, recipe, product, and lifespan category, where applicable.
- The system shall support the agreed export formats.
- Exported data shall respect the user's access permissions and applicable data restrictions.

5. Cross-Functional Business Rules

5.1 Lifespan Accuracy

- The remaining lifespan displayed by the system shall reflect the applicable maximum lifespan and all valid consumption records for the tool.

5.2 Data Consistency

- Tool, machine, recipe, product, consumption, and lifespan information shall remain consistent across the relevant system views and dashboards.

5.3 Access Control

- Users shall perform only the activities permitted for their assigned roles.
- Shift supervisors shall be able to enter missing consumption on behalf of operators under their supervision.

5.4 Traceability

- Consumption entries, supervisor-entered missing records, administrative overrides, bulk operations, and tool replacements shall retain the agreed historical information.
- The system shall identify the user responsible for each recorded action.

5.5 Operational Calendar

- Normal consumption recording shall follow the configured production calendar and shift structure.
- Sunday shall be treated as a normal non-working day, with administrator bulk consumption updates available for selected tools and dates when required.

5.6 Consistent Lifespan Classification

- Healthy, Moderate, and Critical classifications shall use the same approved thresholds throughout the system.

5.7 Replacement History

- Replacing a tool shall establish the replacement tool's new lifespan baseline without unintentionally altering the old tool's historical records.

6. Admin Master Data and System Setup

The administrator shall be responsible for maintaining the master data and operational configurations required for the effective functioning of the Tool Lifetime Management System.

6.1 Machine List Management

- The administrator shall be able to create, view, update, and maintain the machine list.
- Each machine record shall include, at a minimum:
  - Machine ID.
  - Machine location.
- The system shall allow machine information to be used when configuring tool associations and recording consumption.
- The system shall prevent unintended duplicate machine records.
- The system shall support the agreed process for marking machines inactive without unintentionally losing historical records.

6.2 Tool List Management

- The administrator shall be able to create, view, update, and maintain the tool list.

- The tool list shall contain the information required to identify and manage each tool, including:
    - Tool ID or Product Code, as applicable
    - Machine association
    - Recipe number
    - Product code, where applicable
    - Item name
    - Maximum lifespan
    - Tool Cost
    - Tool status
- The system shall validate mandatory information and apply the agreed duplicate-identification rules.
- Changes to tool information shall not unintentionally overwrite historical consumption or lifespan records.

6.3 User List Management

- The administrator shall be able to create, view, update, and maintain the user list.
- User records shall contain the information required to identify users and manage their access, including:
    - User ID or employee ID
    - User name
    - Assigned role
    - Active or inactive status
    - Relevant shift assignment, where applicable
    - Assigned shift supervisor, where applicable
- The system shall support the roles of Operator, Admin, Setter, and Supervisor.
- The administrator shall be able to update user information and role assignments according to the agreed access rules.
- Inactive users shall no longer be able to access the system, while their historical actions shall remain available according to the agreed retention rules.

6.4 Shift Configuration

- The administrator shall be able to configure and maintain the factory's standard shifts.
- The system shall support the three standard production shifts.
- The administrator shall be able to maintain relevant shift information, including:
    - Shift name or identifier.
    - Shift start time.
    - Shift end time.
    - Applicable working days.

    - Shift status, where applicable.
- Configured shifts shall be available for consumption recording and operator shift assignments.
- The system shall apply the configured shift information consistently when recording and reviewing consumption.
- Shift changes, overnight shifts, and exceptional Sunday production shall follow the agreed operational rules.

6.5 Shift Assignment to Operators

- The administrator shall be able to assign operators to their applicable production shifts.
- The system shall allow the administrator to create and update operator shift assignments based on factory requirements.
- Each assignment shall identify the operator and the applicable shift.
- Where required, assignments shall also identify the applicable machine, working days, and assignment validity period.
- The administrator shall be able to update assignments when operators move between shifts or when shift allocations change.
- The system shall use shift assignments to support consumption recording and identification of missing entries.
- Changes to assignments shall not alter historical consumption records.

6.6 Shift Supervisor Details and Assignments

- The administrator shall be able to maintain shift supervisor details and assignments.
- The system shall allow the administrator to associate supervisors with relevant shifts and operator groups.
- The system shall use these assignments to determine which supervisor is authorized to enter missing consumption on behalf of an operator.
- Supervisors shall be able to perform this action only within their assigned scope.
- The administrator shall be able to update supervisor assignments when shifts, operator allocations, or supervisory responsibilities change.
- Changes to supervisor assignments shall not alter the identity of the person responsible for historical consumption entries.

6.7 Excel Upload for Master Data

- The system shall provide Excel upload functionality for the administrator to upload:
    - User list.

  - Machine list.
  - Tool list.
- The upload functionality shall support bulk creation of master records and, where agreed, updates to existing records.

User List Upload

- The administrator shall be able to upload user information, including user identifier, name, role, status, and relevant shift or supervisor assignments where applicable.
- The system shall validate mandatory fields and user identifiers.

Machine List Upload

- The administrator shall be able to upload machine information, including machine ID and location.
- The system shall validate mandatory fields and identify duplicate or invalid machine records.

Tool List Upload

- The administrator shall be able to upload tool information, including the applicable tool identifier, machine association, recipe number, product code where applicable, item name, maximum lifespan, description, and status.
- The system shall validate the uploaded information and relevant master data associations.

Upload Validation and Error Handling

- The system shall validate uploaded files against the agreed Excel template and required column structure.
- The system shall identify missing mandatory fields, invalid values, and duplicate identifiers.
- The system shall validate references to existing master records where applicable.
- The system shall provide an upload result indicating successful records and records requiring correction.
- The system shall identify the relevant row and reason for each rejected record.
- The system shall prevent invalid records from being created.
- The system shall avoid unintentionally overwriting existing records or historical information.
- The administrator shall be able to correct rejected records and upload them again.

- The handling of files containing both valid and invalid records, including whether valid rows can be imported independently, shall follow the agreed upload rules.

6.8 Master Data Consistency

- The administrator shall be responsible for keeping machine, tool, user, shift, operator assignment, and supervisor information accurate and up to date.
- The system shall validate relevant master data associations before allowing related operational activities to be completed.
- Master data changes shall not unintentionally alter historical consumption, lifespan, or user activity records.
- Current user assignments and permissions shall be applied consistently across consumption recording, missed-entry handling, administrative operations, and dashboard access.